very high masking rates. This phenomenon can be attributed to two counteracting effects. Firstly, the masking process reduces the number of active edges, potentially leading to a decrease in overall model capacity. Secondly, the remaining connections become less redundant due to the sparsity introduced by masking. This may allow individual connections to exert a more significant influence on the model's robustness. Meanwhile, as we increase the masking rate, we see in Figure 14 that the negative values of ORC diminish. Furthermore, the ORC distributions of the final attention layer exhibit a convergence towards modal values across all examined datasets, which suggests a diminishing extremity within these distributions. This provides another piece of evidence for our claims on the connection between ORC in the attention map and the robustness of the transformer.

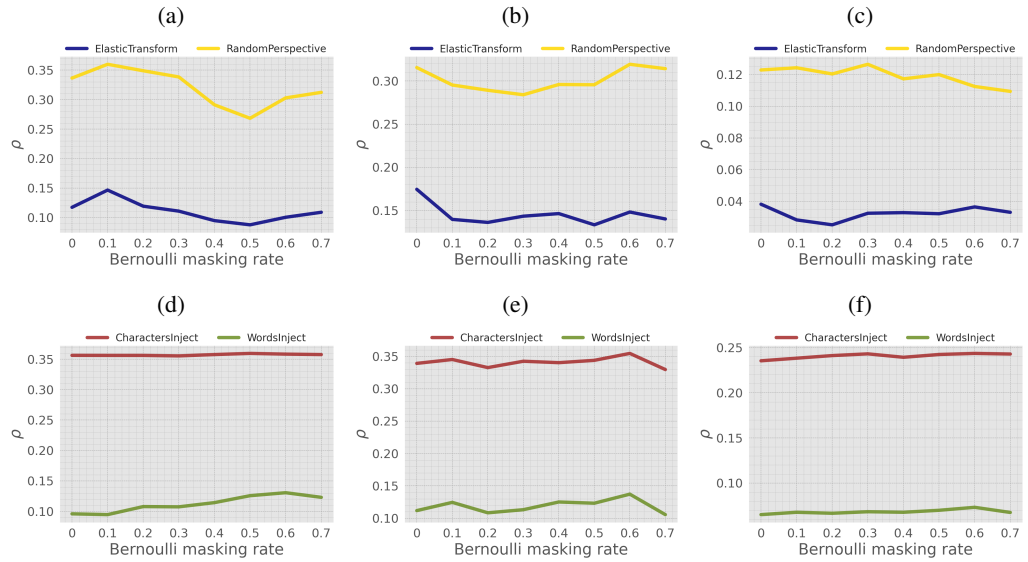

Figure 13: Perturbed test set accuracy rate degradation (ρ as defined in Equation 8) of *L-ViT* (top row) and BERT-Tiny (bottom row) models trained with Bernoulli masking. Results are evaluated over random subsets of the: (a) MNIST; (b) Fashion-MNIST; (c) CIFAR10; (d) IMDb; (e) SST-2; and (f) Headlines datasets. Masking rates affect model robustness in both vision and language models.

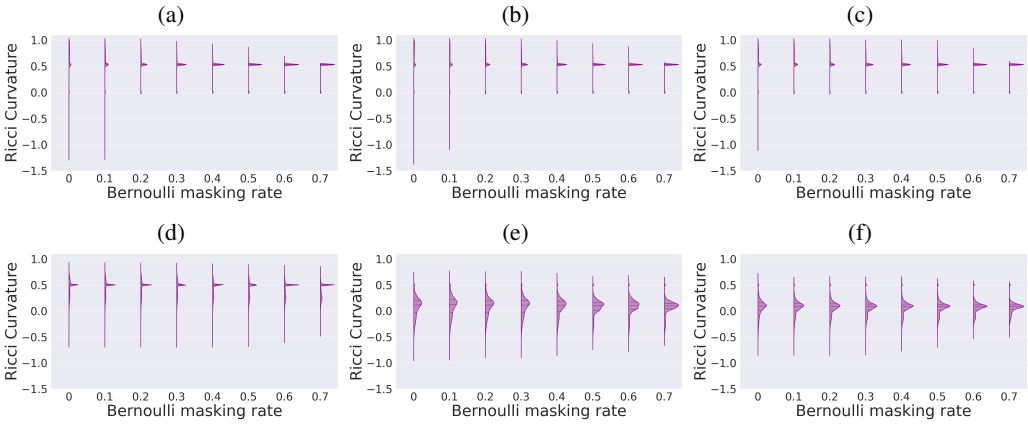

Figure 14: ORC distributions of the final layers of *L-ViT* (top row) and BERT-Tiny (bottom row) models trained with varying edge masking rates. Distributions are combined over random subsets of the: (a) MNIST; (b) Fashion-MNIST; (c) CIFAR10; (d) IMDb; (e) SST-2; and (f) Headlines datasets. Higher edge masking rates lead to curvature values convergence to a common modal value. Consequently, negative curvature values tend to diminish in both vision and language models.

F RELATED WORK

In this work, we investigate the connections between the geometry of attention maps in transformers, their training, performance, and robustness properties. While related work in transformers, geometric deep learning, and robustness is referenced throughout the paper where relevant, we now provide additional discussion of the most related works in each area here.

Transformers and attention mechanism. Transformers (Vaswani et al., 2017) have revolutionized deep learning, achieving state-of-the-art performance across various domains (Brown et al., 2020; Devlin et al., 2018; Dosovitskiy et al., 2020). Recent works have begun to shed light on their inner workings, providing insights into their expressive power (Zaheer et al., 2020), generalization properties (Li et al., 2023), in-context learning (Von Oswald et al., 2023), learning dynamics (Abbe et al., 2024; Dovonon et al., 2024), and robustness characteristics (Bhojanapalli et al., 2021; Shao et al., 2021). However, many questions regarding the fundamental principles governing the behavior of transformers remain unanswered. Our work contributes to this growing body of research by investigating the connection between the geometry of attention maps and the properties of transformers.

Graph Ricci curvature. Ricci curvature captures the deviation of the local geometry from Euclidean space (Bauer et al., 2017). When extended to discrete structures like graphs, it has proven to be a powerful tool for various deep learning tasks (Di Giovanni et al., 2023; Liu et al., 2023; Southern et al., 2023; Topping et al., 2021). We employ the Ollivier-Ricci curvature (ORC) (Ollivier, 2009), an optimal transport formulation of Ricci curvature on graphs, to study the geometry of attention maps in transformers. To the best of our knowledge, this is the first work to explore this connection.

Curvature and robustness. Ricci curvature has been linked to the robustness of systems (Pouryahya et al., 2017), owing to its relationship with entropy. This connection is highlighted by a fundamental result from optimal transport theory (Lott & Villani, 2009), which provides bounds on entropy based on a lower bound of the Ricci curvature. The Fluctuation Theorem (Evans et al., 1993) further establishes a positive correlation between the robustness and entropy of a system, implying that robustness and curvature are positively correlated (Pouryahya et al., 2017). Leveraging this connection, we explore the implications of the ORC distribution for the robustness of vision and language transformers. In summary, our key contribution is uncovering insightful patterns in the curvature distribution of attention maps that evolve during training and through the layers of transformers, establishing a connection between the geometry of attention and the properties of these models. To the best of our knowledge, this is the first work to explore this connection, opening up new avenues for developing more robust and interpretable transformers.